Air Compressor Assembly

Service and Diagnostic Manual
Class 8 Air Brake System Platform

Part Number: ACM-2800
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Section 1: General Description

The ACM-2800 Air Compressor Assembly provides the compressed air supply for the vehicle air brake system, suspension air management, and auxiliary pneumatic accessories. The compressor is engine-driven through a gear train coupled to the engine timing gears.

Section 2: Component Identification

The ACM-2800 assembly consists of seven sub-component groups described in Table 2-1.

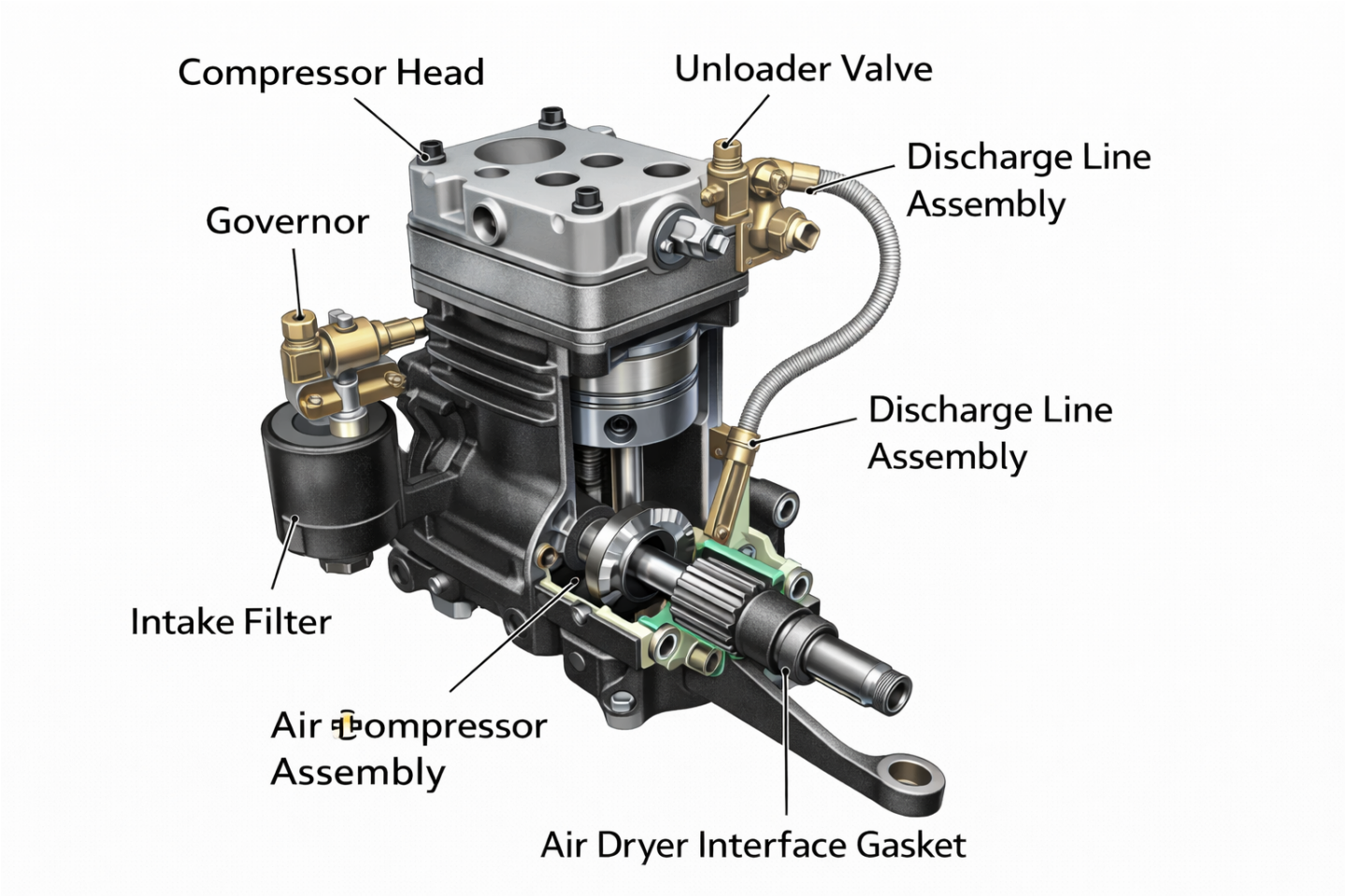

Figure 2-1: ACM-2800 Air Compressor Assembly - Component Layout

Table 2-1: Sub-Component Identification

Item	Sub-Component	Description
1	Compressor Head	Upper cylinder enclosure containing the valve plate and reed valves that control intake and discharge air flow direction during the compression cycle.
2	Piston and Cylinder Kit	Primary compression element. Single-cylinder reciprocating assembly with two compression rings and one oil control ring. Displacement: 18.7 cubic inches per revolution.
3	Unloader Valve	Pneumatic valve that holds inlet reed valves open during unloaded operation, preventing pressure build when system is at cut-out pressure. Controlled by governor signal.
4	Governor	Pressure-sensing regulator monitoring system air pressure via wet tank sensing line. Signals unloader to engage at 130 PSI cut-out and disengage at 100 PSI cut-in.
5	Intake Filter	Cellulose/synthetic blend media rated at 10 micron filtration. Prevents airborne contaminants from entering the compression chamber and accelerating ring wear.
6	Discharge Line Assembly	High-pressure rated steel line routing compressed air from compressor head to air dryer and wet tank. Working pressure rating: 150 PSI.
7	Air Dryer Interface Gasket	Composite sealing gasket at the discharge line-to-air dryer connection. Prevents compressed air leakage at the system entry point to the drying circuit.

Section 3: Specifications

The following specifications apply to all ACM-2800 configurations.

Displacement	18.7 cubic inches per revolution
Maximum Pressure	150 PSI (governed to 130 PSI cut-out)
Cut-in Pressure	100 PSI +/- 3 PSI
Cut-out Pressure	130 PSI +/- 3 PSI
Build-up Time (90 to 120 PSI)	< 40 seconds at 1200 RPM
Cooling Method	Liquid-cooled (engine coolant circuit)
Lubrication	Engine oil pressure-fed
Maximum Oil Carry-over	28 ml per hour at rated speed
Intake Filter Media	Cellulose/synthetic blend, 10 micron
Discharge Line Working Pressure	150 PSI
Mounting Bolt Torque	38 ft-lbs (M12 bolts)
Assembly Weight	42 lbs

Section 4: Installation and Removal

Drain the air system completely before beginning compressor work. Open all drain valves on wet tank, primary tank, secondary tank, and auxiliary tanks. Disconnect discharge line, coolant hoses, oil supply line, and unloader valve air line from governor. Remove four M12 mounting bolts. During installation, apply assembly lubricant to drive gear teeth, align timing marks, and torque mounting bolts to 38 ft-lbs in a cross pattern.

Section 5: Theory of Operation

Compression Cycle

The ACM-2800 is a single-cylinder reciprocating compressor driven at one-half engine crankshaft speed. During the downstroke, the piston draws ambient air through the intake filter and past the inlet reed valve. During the upstroke, air is compressed and forced past the discharge reed valve into the discharge line assembly.

Load/Unload Control

The governor monitors system pressure through a sensing line connected to the wet tank. At cut-out (130 PSI), the governor signals the unloader valve to hold the inlet reed valves open, preventing pressure build. At cut-in (100 PSI), the governor releases the unloader, allowing the compressor to resume building pressure.

Section 6: Troubleshooting and Diagnostics

6.1 Slow Air Pressure Build-up

If build-up time from 90 to 120 PSI exceeds 40 seconds at 1,200 RPM, the most common cause is progressive wear of the piston and cylinder kit compression rings. As the rings wear, the seal between piston and cylinder wall degrades, allowing compressed air blow-by during the upstroke. This reduces effective displacement and increases build-up time. Simultaneously, the worn oil control ring allows engine oil to enter the compression chamber, resulting in elevated oil carry-over into the air system. Technicians may observe that slow build-up is accompanied by oily residue in the air dryer discharge - this combination strongly suggests piston ring wear.

6.2 Excessive Compressor Cycling

If the compressor cycles between loaded and unloaded states every 30 to 90 seconds during steady highway driving with no brake usage, a system air leak is consuming air faster than the compressor can maintain pressure margin. If no external leaks are found, the governor may have an internal pressure sensing leak. A leaking governor reference chamber underestimates system pressure and commands the unloader to disengage prematurely, creating rapid load-unload cycling visible as quick oscillation between cut-in and cut-out on the dash gauge.

6.3 Hissing at Compressor Head

An audible hissing from the compressor head area present only during the loaded (compression) phase indicates failure of the compressor head gasket or reed valve. If the head gasket fails, compressed air escapes between sealing surfaces during each stroke. A cracked reed valve allows backflow from the discharge line assembly into the compression chamber. Perform a leakdown test to differentiate: pressurize through the intake port - air escaping between head and cylinder body is gasket failure; air escaping through the discharge port is reed valve failure.

6.4 Oil Contamination in Air System

Expelled liquid from air tank drain valves that has an oily sheen or milky appearance indicates elevated oil carry-over from the piston and cylinder kit. The excess oil contaminates the air dryer desiccant, which loses moisture-absorbing capacity and allows water vapor to pass through to the air system. The air dryer interface gasket may also show signs of oil saturation and swelling. In cold weather, accumulated moisture freezes in brake system components, potentially causing partial or complete loss of braking.

6.5 Unloader Valve Sluggishness

A slow-to-engage unloader valve allows system pressure to overshoot the cut-out setting before unloading occurs. Conversely, a slow-to-disengage unloader keeps the compressor unloaded longer than necessary after pressure drops below cut-in, extending recovery time and potentially causing low-air-pressure warnings during heavy brake usage such as mountain descents. Internal friction or corrosion from moisture exposure in the unloader mechanism is the typical root cause. Replace the intake filter at recommended intervals to minimize moisture ingress that accelerates unloader corrosion.

Section 7: Preventive Maintenance

At each PM interval, drain all air system reservoirs and note the character of expelled liquid. At each B-service, measure compressor build-up time and record it. Replace the intake filter at the manufacturer's interval or more frequently in dusty environments. At 250,000 miles, measure oil carry-over rate by collecting air dryer discharge effluent. If rate exceeds 40 ml/hr, proactively replace the piston and cylinder kit.